

Methods: Spatial Distribution Analysis

1 Methods: Spatial Distribution Analysis via Morisita's Index of Dispersion

To quantify the spatial distribution of fluorescent particles in **membranes**, Morisita's Index of Dispersion (I_M) was computed using a quadrat-based random sampling approach. The complete analysis pipeline is described below.

- **Image Pre-processing :** Fluorescence microscopy images acquired under varying buffer conditions (NaHEPES and Naphosphate, at concentrations of 10, 50, 100, and 250 mM, across pH values of 6.0, 6.5, 7.0, 7.5, and 8.0) were imported in grayscale format using the OpenCV library. Each image was represented as a two-dimensional array of pixel intensity values in the range $[0, 255]$.
- **Image Binarization :** Prior to spatial analysis, each grayscale image was converted to a binary mask to isolate fluorescent signal from background. Otsu's automatic thresholding method [2] was applied, which determines an optimal global intensity threshold by minimising the intra-class variance between foreground and background pixel populations. Pixels with intensity values exceeding the Otsu threshold were assigned a value of 1 (bright/signal), while all remaining pixels were assigned a value of 0 (background). This approach was adopted to ensure that binarization was objective and reproducible

across all experimental conditions, without requiring manual threshold selection.

- **Random Quadrant Sampling :** To assess spatial heterogeneity, $q = 1000$ non-deterministic square quadrats of side length $s = 5$ pixels were randomly positioned within the binarized image. For each quadrat Q_i , the number of bright pixels n_i was recorded by summing all foreground pixels within the corresponding $s \times s$ pixel region of the binarized image.
- **Computation of Morisita's Index of Dispersion :** Morisita's Index of Dispersion (I_M) was computed from the quadrat counts following the formulation originally proposed by Morisita [1]:

$$I_M = \frac{q \sum_{i=1}^q n_i (n_i - 1)}{N (N - 1)} \quad (1)$$

where $N = \sum_{i=1}^q n_i$ is the total number of bright pixels observed across all q sampled quadrats. The index I_M provides a dimensionless measure of spatial aggregation, the signification of which can be obtained using a hypothesis The testing framework is explained next.

- **Statistical Significance Testing :** The statistical significance of the observed I_M value was assessed using the chi-squared test derived by Morisita [1]. Under the null hypothesis H_0 that the bright pixels are spatially randomly distributed ($I_M = 1$), the test statistic is given by:

$$\chi^2 = I_M (N - 1) + q - N \quad (2)$$

with degrees of freedom $df = q - 1$. The corresponding p -value was obtained from the chi-squared cumulative distribution function [3]. A significance threshold of $\alpha = 0.05$

was applied: if $p < 0.05$, the null hypothesis was rejected and the distribution was classified as either *clustered* ($I_M > 1$) or *uniform* ($I_M < 1$); if $p \geq 0.05$, the distribution was classified as *random*.

References

- [1] Morisita, M. (1959). Measuring the dispersion of individuals and analysis of the distributional patterns. *Memoirs of the Faculty of Science, Kyushu University, Series E (Biology)*, 2, 215–235.
- [2] Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62–66.
- [3] Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272.